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Strains of bacteria with antibiotic resistance are common in hospitals.

Use your knowledge of selection to explain why.

[3 marks]

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Antibiotic resistance alleles are located in plasmids.

There are 11 different types of plasmid inside cells of the bacterium
Bacillus thuringiensis.

A scientist determined the:

- length of each type of plasmid in kilobase pairs (kbp)
- mean plasmid copy number (mean number of each type of plasmid per cell).

Table 5 shows the scientist's results for 5 of these plasmids.

Table 5

Plasmid identity	Plasmid length / kbp	Mean plasmid copy number	Rank of plasmid length	Rank of mean plasmid copy number	Squared difference between ranks (d^2)
A	8	172	2	5	9
B	95	6	4	2	4
C	12	35	3	3	0
D	2	139	1	4	9
E	294	3	5	1	16

The Spearman rank correlation coefficient (r_s) between plasmid length and mean plasmid copy number is determined using this equation.

$$r_s = 1 - \frac{6 \sum d^2}{n(n^2 - 1)}$$

where

$\sum d^2$ = the sum of all d^2 figures

$n = 5$ (the highest rank number)

Use information in **Table 5** and the formula to calculate the r_s value for this set of data.

Show your working.

[2 marks]

$r_s =$ _____

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The scientist determined that the r_s value between plasmid length and mean plasmid copy number for all 11 types of plasmid is -0.94

What can you conclude from this r_s value?

[1 mark]

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The length of the single circular (non-plasmid) DNA molecule in a *B. thuringiensis* cell is 5400 kbp

The total length of all plasmid DNA molecules in a *B. thuringiensis* cell is 65% greater than the length of the single circular DNA molecule.

Calculate the total number of kbp in all the plasmid molecules in a *B. thuringiensis* cell.

Show your working.

[2 marks]

Answer _____ kbp

Question 8 continues on the next page

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ANSWER IN THE SPACES PROVIDED**

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08.5

A growing population of *B. thuringiensis* cells includes a:

- germination phase
- growth phase
- dormant phase.

A scientist incubated a culture of *B. thuringiensis* cells at 20 °C

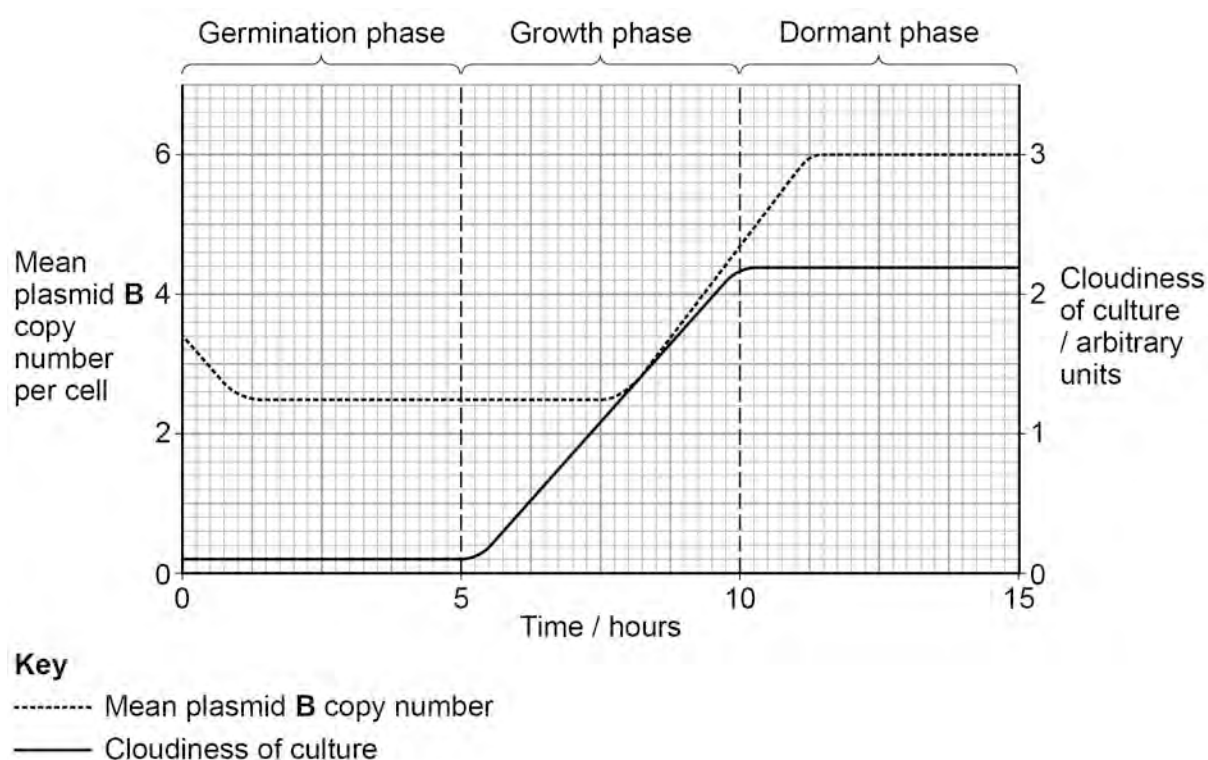
At intervals, the scientist:

- measured the cloudiness of the culture
- determined the mean plasmid **B** copy number per cell.

Cloudiness is a measure of the number of cells in a culture.

Figure 11 shows the scientist's results.

Figure 11



Using information in **Figure 11**, what can you conclude about binary fission and the replication of plasmid **B** in the different phases of a growing population of *B. thuringiensis*?

[3 marks]

1

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